

Design Calculation Sheet

Case ID: t3_r1_mimo
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1. Capacity Calculation

Parameter	Value	Source
Nominal cell capacity	2.3077 Ah	r4_enhanced_1c.json:capacity_ah
1C current	2.3077 A	I = capacity × 1C
5C current	11.539 A	I = capacity × 5C
5C discharge capacity	2.2345 Ah	r4_enhanced_5c.json:capacity_ah
5C retention	**96.83%**	2.2345 / 2.3077

2. Energy Density Calculation (Contract Caliber)

Parameter	Value	Formula	Source
Discharge energy	8.339 Wh	$\int V \cdot I dt / 3600$	r4_energy.json:energy_wh
Cell mass (excl. electrolyte)	0.02704 kg	$\Sigma \text{ layer thickness} \times (1 - \epsilon) \times \rho \times \text{area}$	r4_energy.json:mass_kg
Gravimetric ED	**308.4 Wh/kg**	Energy / mass	r4_energy.json:energy_density_wh_kg
Cell volume	$1.171 \times 10^{-4} \text{ m}^3$	$\Sigma \text{ layer thickness} \times \text{area}$	r4_energy.json:volume_m3
Volumetric ED	**712.3 Wh/L**	Energy / volume	r4_energy.json:energy_density_wh_l

3. Power Density Calculation

Parameter	Value	Formula	Source
OCV (midpoint)	3.845 V	Voltage at discharge midpoint	r4_energy.json:midpoint_voltage_v
DC Resistance	2.35 mΩ	$(V_{\text{start}} - V_{10\%}) / I$	r4_energy.json:dcr_ohm
Max power (Ragone)	$V^2 / (4 \cdot \text{DCR})$	Theoretical max	Derived
Power density	**65,280 W/kg**	Max power / mass	r4_energy.json:power_density_w_kg

4. Thermal Analysis

Parameter	Value	Source
T_max (1C discharge)	300.2 K (27.1°C)	r4_enhanced_1c.json:T_max_K
T_max (5C discharge)	316.0 K (42.9°C)	r4_enhanced_5c.json:T_max_K
T_max (4C charge at 45°C)	327.6 K (54.5°C)	r4_enhanced_4c.json:T_max_K
Thermal limit	**333.15 K (60°C)**	Task criterion
Thermal margin	**5.5 K**	333.15 – 327.6

5. Plating Analysis

Parameter	Value	Criterion	Status
Anode potential minimum	+0.005 V	≥ 0 V (no plating)	✓ PASS
Plating determination	No plating	—	✓ SAFE

6. Layer Thickness Budget

Layer	Thickness (μm)	Cumulative (μm)
Positive CC (Al)	16	16
Positive electrode	35	51
Separator	12	63
Negative electrode	39	102
Negative CC (Cu)	12	114
Total	**114 μm**	—

7. Key Design Ratios

Ratio	Value	Notes
Pos electrode thickness / Neg electrode	0.897	Designed for N/P $\approx$ 1.1
Pos porosity	0.42	High for rate capability
Neg porosity	0.32	Moderate
Separator porosity	0.50	High for electrolyte access
Pos particle radius	2.0 μm	Reduced from 5.22 μm baseline
Neg particle radius	2.0 μm	Reduced from 5.86 μm baseline